

CS677-Spring2025 CRN#22550: Machine Learning

Assignment#3 – Total Points: 100

Due Date: 04/24/2025 (April 24th)

(Team Assignment)

Sr.No	Questions	Points	Self-Assessment																				
1	Try changing the combination and values of hyperparameters for the example mentioned in the provided reference link and explain the impact that it has. https://scikit-learn.org/stable/auto_examples/ensemble/plot_gradient_boosting_regression.html#sphx-glr-auto-examples-ensemble-plot-gradient-boosting-regression-py	20																					
2	Try different values of cross val predict, cv for the example mentioned in the provided reference link and explain the impact that it has. https://scikit-learn.org/stable/auto_examples/model_selection/plot_cv_predict.html#sphx-glr-auto-examples-model-selection-plot-cv-predict-py	20																					
3	Write a Python script that computes the MAE, MSE, RMSE, R-squared, and adjusted R-squared metrics for a given dataset. Analyze the resulting metric values to determine whether the model performed well or not. <table><thead><tr><th>Actual Value</th><th>Predicted value</th></tr></thead><tbody><tr><td>19</td><td>20</td></tr><tr><td>30</td><td>26</td></tr><tr><td>89</td><td>80</td></tr><tr><td>38</td><td>40</td></tr><tr><td>46</td><td>46</td></tr><tr><td>77</td><td>74</td></tr><tr><td>65</td><td>68</td></tr><tr><td>54</td><td>54</td></tr><tr><td>45</td><td>38</td></tr></tbody></table>	Actual Value	Predicted value	19	20	30	26	89	80	38	40	46	46	77	74	65	68	54	54	45	38	20	
Actual Value	Predicted value																						
19	20																						
30	26																						
89	80																						
38	40																						
46	46																						
77	74																						
65	68																						
54	54																						
45	38																						

4	What is the difference between Ridge regression and Lasso regression? How do these two regularization techniques help in preventing overfitting in a regression model? Additional - Can you provide an example where Ridge regression may be more appropriate to use than Lasso regression, and vice versa?	20	
5	How does Gradient Descent, Stochastic Gradient Descent, Mini-Batch Gradient Descent differ in terms of their approach to optimizing the objective function? How does the choice of learning rate affect the convergence of these algorithms to the global or local minima? Is it always the case that the loss function will be decreasing after a parameter update while applying/performing the Stochastic Gradient Descent (SGD) optimizer on a complete data set (no mini batches)? Justify your answer	20	

Submission:

- Submit a Word document with your names and a detailed explanation and draw diagrams wherever necessary. You can also write and draw on paper, and scan to submit (you can submit clear scanned pages).
- Any one Team member can submit.
- Do not submit a zip file.
- You may not be graded full points if your submissions are not clear or unreadable.
- After the **Due Date**, late submission of up to one week will incur a 10% loss of total points earned. 5% every week thereafter until the **end date**.
- Attach this file with self-assessment. This is for your reference if you answered the question completely.